

App Development Project Report

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1. Student Details

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About Me: I am a dual-degree student pursuing a B.Tech in Computer Science and Engineering (third year) alongside the B.S. Degree in Data Science and Applications at IIT Madras, where I am currently in the fourth-term Diploma level.

2. Project Details

Project Title: Trekking Management Application

Problem Statement:

Adventure organizations require efficient systems to manage trekking activities involving trek organizers, staff, and participants. Currently, many trekking groups rely on spreadsheets, phone calls, or manual coordination, which makes it difficult to manage trek approvals, track bookings, avoid overbooking, and maintain trek history.

Approach:

The app was built using Flask and SQLite as the backend framework with a modular blueprint architecture and Jinja2 templating. It uses role based access for admins to oversee operations, staff to manage assigned trek capacities and users to browse and book slots.

3. AI/LLM Declaration

I used GEMINI to assist with debugging code, explaining programming errors, optimizing database queries, and refining certain backend route logic during the development of this project. The extent of AI/LLM usage was approximately 15–20%, primarily for error resolution,

code suggestions, and technical explanations. All final implementation, integration, testing, and project decisions were completed manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Flask-Login	User authentication and session management
SQLite	Lightweight local database for storing user data

5. Database Schema / ER Diagram

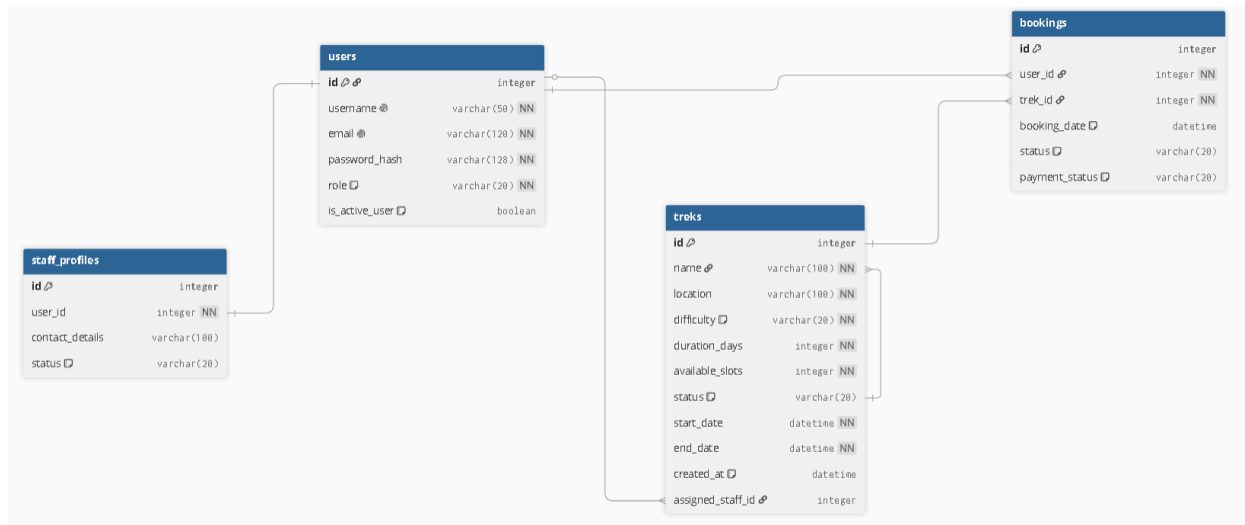
Tables:

1. **User** — stores user credentials and roles (id, username, email,)
2. **Activity** — logs user activities (id, user_id, activity_type, duration, date, mood)
3. **ActivityType** — stores predefined activity categories (id, name, description)

Relationships:

- One-to-Many → **User** → **Staff Profile**
- One-to-Many → **User** → **Trek**

This is my ER Diagram



6. Architecture and Features (optional)

Architecture Overview:

- **app.py** – main Flask application entry point and auto - database installation
- **/models** – database models using SQLAlchemy
- **/routes** – Flask Blueprints for admin, staff, trekker, and authentication routes
- **/templates** – Jinja2 HTML templates
- **/static** – CSS

7. Video Presentation

Drive Link:

[Mad1 demo video](#)